Experiment 11: Water Jug Problem

Aim

To solve the Water Jug Problem using Python.

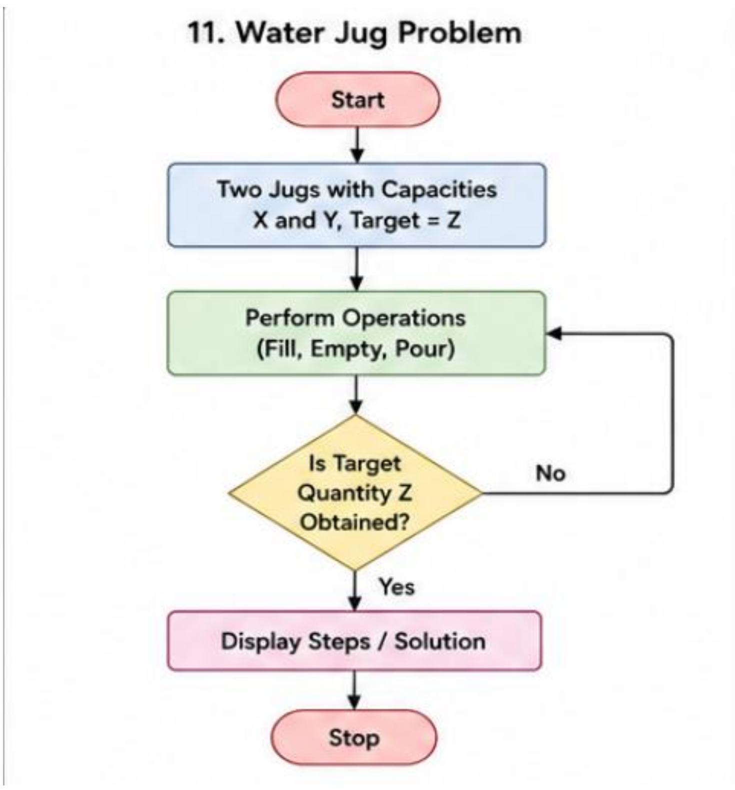
Objective

To measure the required quantity of water using two jugs of different capacities.

Algorithm

1. Define the capacities of the two jugs.
2. Fill the larger jug.
3. Pour water from one jug to another.
4. Empty or refill the jugs when required.
5. Repeat until the required amount of water is obtained.
6. Display the steps.

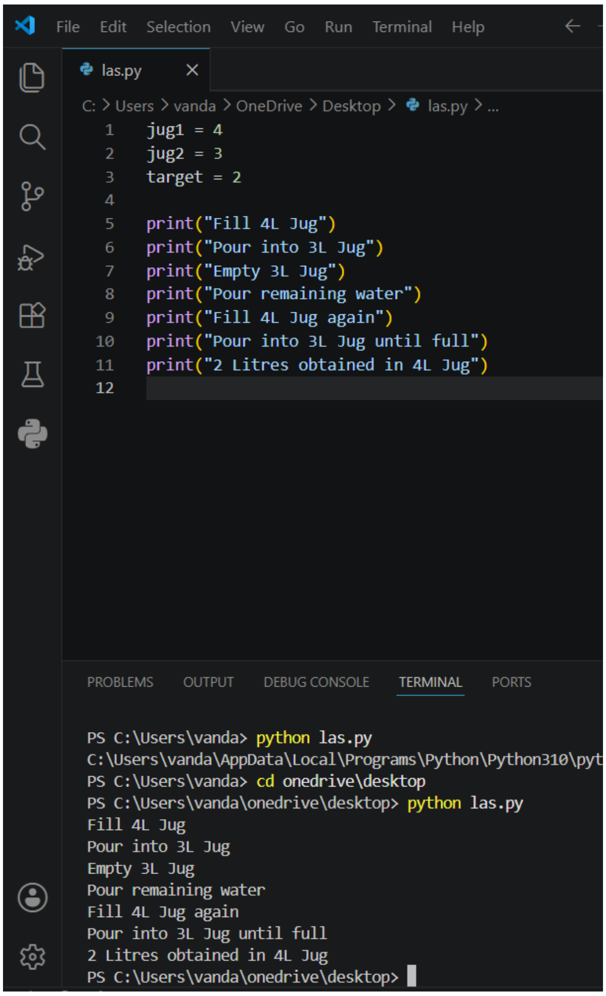
Flowchart



Code

```
jug1 = 4

jug2 = 3

target = 2


print("Fill 4L Jug")

print("Pour into 3L Jug")

print("Empty 3L Jug")

print("Pour remaining water")

print("Fill 4L Jug again")

print("Pour into 3L Jug until full")

print("2 Litres obtained in 4L Jug")
```

Output :



Result

The required quantity of 2 litres of water was obtained successfully.

Conclusion

The Water Jug Problem was solved successfully using a sequence of fill, pour, and empty operations. It demonstrates the concept of state-space search in Artificial Intelligence.